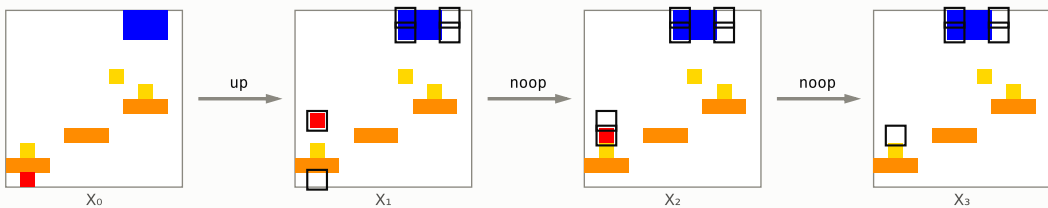


Mario (N2NTD) — three transitions from one training drive



$X_0 \rightarrow X_1$ **up** the red player jumps 4 rows, (11,1) $\rightarrow$ (7,1), straight through the platform above it
 $X_1 \rightarrow X_2$ **noop** no action, and it falls back one row
 $X_2 \rightarrow X_3$ **noop** it falls onto the coin at (9,1) and disappears — the player is drawn underneath the gold
 on every one of these steps the blue 2×3 block also marches one column left, whatever the agent does

	abstraction program P	world knowledge K
node 0 seed 0.00 min(ID, cFD)	blank slate — parses the grid, emits nothing <pre> grid = _parse_grid(obs) ... return "" P(X₁) = (empty string) P(X₂): identical — the transition is invisible to P </pre>	(empty — no world knowledge yet)
node 2 iter 3 · P 0.23 min(ID, cFD)	updated — one flat list of non-background cells <pre> parts_alt = [f"grid {rows}x{cols}; bg:{bg}; cells:"] for r, c, clr in cells: parts_alt.append(f"({r},{c},{clr});") summary = "".join(parts_alt) P(X₁) = grid 12x12; bg:white; cells:(0,7,blue);(0,8,blue);(0,9,blue);(1,7,blue);(1, 8,blue);(1,9,blue);(4,7,gold);(5,9,gold);(6,8,darko range);(6,9,darkorange);(6,10,darkorange);(7,1,red); (8,4,darkorange);(8,5,darkorange);(8,6,darkorange);(9,1,gold);(10,0,darkorange);(10,1,darkorange);(10,2, darkorange) P(X₂): -cells:(0,7,blue) -(0,9,blue) -(1,9,blue) -(7,1,red) +cells:(0,6,blue) +(0,7,blue) +(1,6,blue) +(8,1,red) </pre>	(empty — no world knowledge yet)
node 3 iter 4 · K 0.69 min(ID, cFD)	unchanged from node 2	new - The world contains a red cell that is moved by directional actions: ... up ... moves it 4 cells upward ... left ... moves it 1 cell left ... right ... moves it 1 cell right - The blue 2×3 block moves autonomously: each time step it shifts either left or right by 1 column. This movement is independent of the action taken - All other objects (gold cells, dark orange L-shaped groups) are stationary and never change ↑ and no gravity rule at all — refuted by the transitions above
node 16 iter 20 · K 0.79 min(ID, cFD)	inherited from node 15 — grouped by colour <pre> cells_by_colour.setdefault(colour, []).append((r, c)) ... coords_str = " ".join(f"({r},{c})" for r, c in positions_sorted) summary_parts.append(f"{colour}: {coords_str}") P(X₁) = grid 12x12; bg:white; blue: 0,7 0,8 0,9 1,7 1,8 1,9 darkorange: 6,8 6,9 6,10 8,4 8,5 8,6 10,0 10,1 10,2 gold: 4,7 5,9 9,1 red: 7,1 P(X₂): -blue:0,9 -blue:1,9 -red:7,1 +blue:0,6 +blue:1,6 +red:8,1 </pre>	updated - The red cell moves downward by 1 cell on each noop step, unless it is at row 11 or the cell directly below is occupied by any other object - When a red cell moves ... onto a cell containing a gold cell, the gold cell disappears immediately. The red cell does not appear in that cell until the next time step - It moves right until its rightmost column reaches 11, then moves left until its leftmost column reaches 0 ... oscillating continuously